

EXPERIMENT (9)

Ex. No.: 9

Date:

AIM

To perform JOIN operations such as INNER JOIN, LEFT JOIN, RIGHT JOIN and CROSS JOIN on the Hostel Management System tables.

DESCRIPTION

A JOIN combines rows from two or more tables using a related column. MySQL supports CROSS JOIN, INNER JOIN, LEFT JOIN and RIGHT JOIN.

SYNTAX

CROSS JOIN

```
SELECT * FROM table1 CROSS JOIN table2;
```

INNER JOIN

```
SELECT * FROM table1 INNER JOIN table2 ON table1.col=table2.col;
```

LEFT JOIN

```
SELECT * FROM table1 LEFT JOIN table2 ON table1.col=table2.col;
```

RIGHT JOIN

```
SELECT * FROM table1 RIGHT JOIN table2 ON table1.col=table2.col;
```

Questions and Answers

1. List all students along with their room details (INNER JOIN).

```
SELECT s.StudentID, s.StudentName, r.RoomNumber, r.BlockName  
FROM Hostel_Student s  
INNER JOIN Hostel_Rooms r  
ON s.RoomID = r.RoomID;
```

SQL File 3* SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* x

Limit to 1000 rows

```

167      s.StudentName,
168      s.Gender,
169      s.MobileNo,
170      r.RoomNumber,
171      r.BlockName,
172      r.RoomType
173 FROM Hostel_Student s
174 INNER JOIN Hostel_Rooms r
175 ON s.RoomID = r.RoomID;

```

Result Grid

StudentID	StudentName	Gender	MobileNo	RoomNumber	BlockName	RoomType
201	Lokesh	M	9876543210	A101	A Block	Single
202	Rahul	M	9876543211	A102	A Block	Double

en 721 Hostel_Rooms 722 Result 723 Hostel_Warden 724 Hostel_Rooms 725 Result 726 Read Only

2. Display all rooms and the students staying in them, including rooms with no students (LEFT JOIN).

```

SELECT r.RoomID, r.RoomNumber, s.StudentName
FROM Hostel_Rooms r
LEFT JOIN Hostel_Student s
ON r.RoomID = s.RoomID;

```

SQL File 3* SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* x

Limit to 1000 rows

```

175 ON s.RoomID = r.RoomID;
176 • SELECT r.RoomID,
177      r.RoomNumber,
178      r.BlockName,
179      s.StudentID,
180      s.StudentName
181 FROM Hostel_Rooms r
182 LEFT JOIN Hostel_Student s
183 ON r.RoomID = s.RoomID;

```

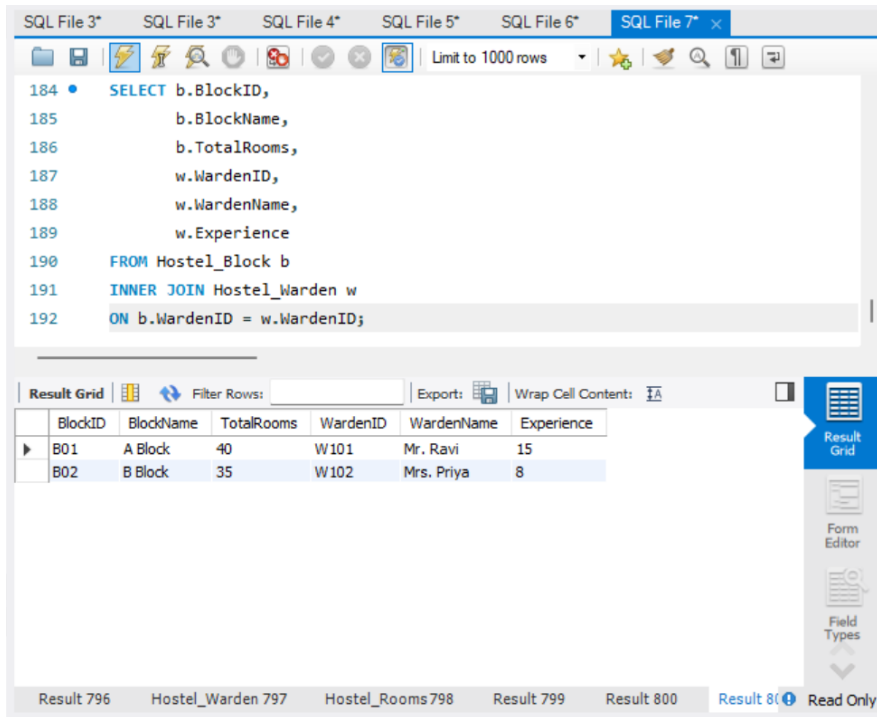
Result Grid

RoomID	RoomNumber	BlockName	StudentID	StudentName
101	A101	A Block	201	Lokesh
102	A102	A Block	202	Rahul
103	B201	B Block	NULL	NULL
104	A104	A Block	NULL	NULL
105	A105	A Block	NULL	NULL
106	B202	B Block	NULL	NULL
107	B203	B Block	NULL	NULL

el_Rooms 758 Result 759 Hostel_Warden 760 Hostel_Rooms 761 Result 762 Result 763 Read Only

3. Display all hostel blocks and their wardens (INNER JOIN).

```
SELECT b.BlockName, w.WardenName, w.Experience  
FROM Hostel_Block b  
INNER JOIN Hostel_Warden w  
ON b.WardenID = w.WardenID;
```



The screenshot shows a SQL IDE window with multiple tabs. The active tab, 'SQL File 7*', contains the following SQL query:

```
184 • SELECT b.BlockID,  
185         b.BlockName,  
186         b.TotalRooms,  
187         w.WardenID,  
188         w.WardenName,  
189         w.Experience  
190 FROM Hostel_Block b  
191 INNER JOIN Hostel_Warden w  
192 ON b.WardenID = w.WardenID;
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query. The grid has the following columns: BlockID, BlockName, TotalRooms, WardenID, WardenName, and Experience. The results are as follows:

BlockID	BlockName	TotalRooms	WardenID	WardenName	Experience
B01	A Block	40	W101	Mr. Ravi	15
B02	B Block	35	W102	Mrs. Priya	8

The IDE interface also includes a toolbar with various icons, a 'Limit to 1000 rows' dropdown, and a sidebar with 'Result Grid', 'Form Editor', and 'Field Types' options. The status bar at the bottom indicates 'Result 800' and 'Read Only'.

RESULT

Thus, the records were successfully retrieved using JOIN operations on the Hostel Management System tables.